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IMPLEMENTATION OF THE RECOMMENDATION ON THE ETHICS OF ARTIFICIAL INTELLIGENCE (AI)

OUTLINE

Source: 217 EX/Decision 4.I.C.

Background: In 217 EX/Dec.4.I.C, the Executive Board decided to inform the General Conference at its 42nd session of progress made in the implementation of the Recommendation on the Ethics of Artificial Intelligence.

Purpose: This document presents the report examined by the Executive Board at its 217th session on the implementation of the Recommendation on the Ethics of Artificial Intelligence, the preparation and the piloting of the readiness assessment methodology and the ethical impact assessment, nationally.

Decision required: No decision required.

1. The implementation of the 2021 Recommendation on the Ethics of Artificial Intelligence (AI) (hereinafter referred to as "the Recommendation") has advanced significantly at the national and global level. Working with Member States and partner institutions, UNESCO has mobilized resources and

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provided comprehensive support to a growing number of countries around the world. This has contributed to advancing the necessary elements to achieve robust governance frameworks, in order to promote the beneficial use of this technology and to safeguard the human rights and fundamental freedoms of their citizens. The Group of Friends of the Implementation of the Recommendation, counting over 80 active Member States, has been a source of support.

2. The comprehensive implementation plan of the Recommendation includes multiple mutually reinforcing elements and actions at the global, regional, and national level:

- establishment of a network of experts to secure state of the art expertise on the best way to govern these technologies.
- operationalization of platforms and conferences for peer learning and exchange, including a Global Observatory, and the Global Forum on the Ethics of AI.
- ensuring a multistakeholder approach by direct and structured engagement with the private sector, civil society, and policy-makers, including parliamentarians.
- analytical and diagnostic work at the country level, using the two implementation tools developed by the Secretariat.

3. At the global level, UNESCO is engaging with major players in the private sector, thinktanks and foundations, along with relevant processes to ensure high impact in the definition on the best way to govern these technologies. This includes UNESCO's participation as a Knowledge Partner of the G20 Indian Presidency within the Digital Economy Working Group, with a view to leveraging the "Labor and Economy" and "Education and Research" Policy Action Areas of the Recommendation, as well as an observer in Global Partnership on AI (GPAI) and its sub-entities (the Steering Committee, Council, and the thematic working groups). UNESCO is also a strategic partner of the W20, where it has leveraged the gender chapter of the Recommendation to inform policies aimed at reducing the digital gender divide, and a partner of Hiroshima Process on Artificial Intelligence, launched in the framework of the Japanese Presidency of the G7.

4. UNESCO is reinforcing its role in the global debate on AI by conducting research and producing new analytical outputs. To address the latest developments in the field of AI, with the widespread proliferation of AI foundation models and launch of major Natural Language Programs that are feeding into chat-box developments (ChatGPT), the Secretariat published a policy paper on AI Foundation Models, using as a basis the provisions of the Recommendation to determine the level of safety, transparency and reliability of the new tools and to guide policy responses. It has been recognized by the Interagency group of AI at the United Nations High-Level Committee on Programmes (HLCP) as a very useful document.

5. The Global Forum on the Ethics of AI, with the first edition hosted by Czechia in the context of their Presidency of the European Union and the second edition to be hosted by Slovenia in early 2024, offers an excellent platform to advance peer learning and exchanges in a multistakeholder and multidisciplinary fashion, and to build a robust international coalition ensuring the ethical development and use of AI worldwide.

6. UNESCO is also actively engaged at the regional level to promote a regional approach to the implementation. For example, UNESCO has partnered with the Development Bank in Latin America (CAF) to reinforce the implementation of the Readiness Assessment Methodology (RAM) in Latin America and the Caribbean and which will be discussed at a first Ministerial Meeting in Santiago de Chile on 23 and 24 October 2023. UNESCO has also established the Ibero-American Business Council

for Ethics of AI, to ensure close collaboration with the private sector in this domain. Co-chaired by Microsoft and Telefonica, the Council is committed to strengthening technical capacities in ethics and AI, including through the application of UNESCO's Ethical Impact Assessment (EIA) tool. Discussions are being held to broaden the Council to all the regions.

7. The Southern Africa Sub-regional Forum on AI (SARFAI) was hosted by Namibia in Windhoek in September 2022 and also launched the RAM in Africa. The East African subregional Forum on AI (EARFAI) is being planned to be hosted by Kenya in Nairobi.

8. Shortly after the adoption of the Recommendation, the Director-General of UNESCO stressed the need for legislative, institutional and regulatory changes at the national level for the Recommendation to be translated into action and have a positive impact on the ground. To assist Member States in this regard, and as mandated by the Recommendation, UNESCO has produced and deployed tools that help standardize the analysis and provide comparative elements for countries to produce and fine-tune their national AI strategies. These diagnostic tools are the RAM and EIA. They are now being piloted in over 50 Member States with a focus on Africa and small island developing States (SIDS) and a strong participation from Latin America and the Caribbean region as well. The countries concerned include Antigua and Barbuda, Barbados, Botswana, Brazil, Chad, Chile, Costa Rica, Cuba, Democratic Republic of the Congo, Dominican Republic, Gabon, India, Indonesia, Kenya, Malawi, Maldives, Mauritius, Mexico, Morocco, Mozambique, Namibia, Rwanda, Sao Tome and Principe, Senegal, South Africa, Timor-Leste, Uruguay and Zimbabwe.

9. Prior to the development of the tools, the Secretariat conducted extensive desk research to identify and analyse the existing tools and their basic parameters, including the scale and scope, as well as major gaps, with the aim of optimizing UNESCO's tools. The research identified significant shortfall in the existing international landscape of available tools, including a lack of transparency in terms of criteria used for the evaluation of countries, an absence of important dimensions, values, and principles in the evaluation methodology, and a lack of focus on issues specific to AI technologies, as many are focused on general digital issues. Analysing these gaps helped shape the content of the UNESCO tools, optimize their overall design, and underscore their importance. Particular attention was given to gender participation and the inclusion of a gender perspective in the tools.

10. For the technical support in the development of the capacity-building tools, a High-Level Expert Group (HLEG) was set up, featuring leading experts from diverse cultural and professional backgrounds. The Group met monthly throughout 2022 to discuss and revise with the Secretariat the specific segments of the evolving tools, taking into account written feedback collected in between the meetings. After the finalization of the tools, the HLEG has become a permanent source of advice, having provided inputs to the paper that UNESCO produced on Foundation Models such as ChatGPT through the prism of the UNESCO Recommendation on the Ethics of Artificial Intelligence.

11. The RAM and EIA were also prepared with the support of a survey, launched in August 2022, to facilitate Member States' input as a guidance for the development of the tools. A report synthesizing the desk research and survey findings will be published online as a reference document.

12. The work to apply the RAM is carried out by a national team knowledgeable of the local context, hired and supervised by UNESCO's Secretariat and supported by "AI Experts without Borders" – an offspring of the HLEG, that has become an international network of experts that brings best practices from around the globe. The Women for Ethical AI platform (Women4EthicalAI), launched at the United Nations Commission on the Status of Women in March 2023, held its inaugural meeting at UNESCO Headquarters in April 2023. With a core group of 17 leading women experts in this field, the platform is championing the higher participation of women in the AI world and the ethical development of these technologies beyond the gender issues. Report on the "Gendered Impacts of AI: Policies and

Safeguards to Regulate New Technologies, Mitigate Risks and Protect Rights" was produced for this purpose.

13. The RAM aims to conduct a holistic examination of where a country stands in terms of preparedness to develop and adopt AI ethically, including institutionally, and how agile and robust are the laws and policies of the country for this purpose. The RAM is structured around five dimensions: (1) legal/regulatory, (2) social/cultural, (3) economic, (4) scientific/educational; and (5) technical/ infrastructural.

14. The RAM is currently available online in UNESDOC. It is also linked to the Global Observatory on Ethics of AI, which is designed in partnership with the Alan Turing Institute as a collaborative and information-sharing platform to increase understanding of the challenges and impact of AI and promote peer learning. Countries interested in conducting RAM will be provided with a country-specific link for an online questionnaire. The Global Observatory on the Ethics of AI will host the results of the analytical work from all participating countries, together with the contributions from experts from around the world. The Observatory will be launched in the second half of 2023.

15. The EIA is a tool with versatile use for both public and private entities. An example of its application in the public sphere is the use of EIA by a government entity involved in the procurement of AI systems to systematically assess the technology and check if it is in line with the ethical standards set by the Recommendation. The tool can also be used directly by any company that wants to follow a gold standard and declare that they align with the Recommendation. The tool covers the whole AI lifecycle, and asks questions about the project team, how responsibilities are shared, multistakeholder engagement, classification of the system, quality and representativeness of data uses, and other issues.

16. The EIA is composed of two main parts that strike a balance between procedure and substance. In the first part, scoping, the goal is to understand the basics of the system and lay out some preliminary questions such as whether automation is the best solution for this case. In addition, questions about the project team and the plan to engage different stakeholders ensue. The second part is dedicated to the implementation of the UNESCO principles. For each principle, questions will aim to assess: (1) whether sufficient procedural safeguards have been put in place to ensure this system is compliant with the Recommendation; and (2) the (potential) positive outcomes and adverse impacts that may arise from the procurement and deployment of the system, specific to its intended use.

17. Both tools feature a strong emphasis on gender. For example, the RAM examines the different laws and policies that countries have on enhancing gender equality. The EIA also examines the level of diversity of the teams building the AI systems.

18. The EIA is currently available online, and Member States that would like their procurement agencies to pilot the tools are invited to contact the Social and Human Sciences Sector.

19. Deep and comprehensive engagement with Member States on the implementation of the Recommendation was made possible by funding partnerships, including the Government of Japan project on *supporting Africa and SIDS to benefit from AI technologies while addressing ethical risks*, with 20 beneficiary countries; McGovern Foundation project on *harnessing the power of AI to promote equal opportunities in the digital world*, with four beneficiary countries; and the project on *supporting Member States in implementing UNESCO's Recommendation on the Ethics of AI through innovative tools*, funded by the European Union with 18 beneficiary countries. Furthermore, UNESCO is advancing its partnership with the European Union and the Dutch Authority for Digital Infrastructure, along with members of the Dutch and European Working Groups of Competent Authorities on AI. This collaboration aims to strengthen the capacity of the relevant Dutch authorities to supervise AI and facilitate synergies between the implementation of the Recommendation on Ethics of AI and the European Union AI Act.

20. In addition to funding support, multiple strategic partnerships have contributed to the rapid implementation of the Recommendation, including with Member States (Canada (Quebec), Czechia, Japan) and prominent institutions such as the Alan Turing Institute and Thomson Reuters Foundation. Specifically with the Alan Turing Institute, partnership involved secondment of personnel to strengthen the Secretariat in terms of AI expertise and collaboration. A Letter of Intent was signed between the Social and Human Sciences Sector and the Interparliamentary Forum on Emerging Technologies, to mobilize parliamentarians worldwide in support of the implementation of the Recommendation, including through the production of toolkits for parliamentarians. The Thompson Reuters Foundation will mobilize civil society and journalists to advance the implementation of the Recommendation and ensure that they participate in an inclusive national and international debate.

21. While initial progress has been made through mobilization of over US $5.6 million voluntary contributions, further financial and in-kind support is requested from Member States to continue building the leading role that UNESCO has earned in this space, and that is so timely now that global attention is on finding workable solutions on the governance front, to ensure AI is developed and deployed for good and to advance the full implementation of the Recommendation globally. Additional resources, including specialized human expertise, are needed to reinforce the Secretariat, enabling more knowledge productions in light of latest technological developments; while at the same time, ensuring UNESCO’s continuous and prominent presence in all the relevant international and regional fora. The support will also allow to take a step forward in the implementation of the tools in countries that are currently engaged, and to extend its implementation to a greater number of countries. Moreover, resources will need to be mobilized to facilitate regional mechanisms for the implementation, as well as for the different networks of experts and the Global Observatory on Ethics of AI.

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